MM development architecture · v1.0.2

non-normative explanatory derivative · certified MM authorities control

Superior certified authorities

Scientific Execution Architecture v1.0.2

Software Development & Assurance Manual v1.0.1 RC3

Project Evidence Base v1.0.0

Controlled Methodology v1.0.1
human-readable baseline — non-normative

Execution-control governance

CONTROL PLANE
Gates · scope · permissions · adjudication

EXECUTION PLANE
Explicitly authorized mutable work

AUTHORITY PLANE
Immutable identities · no implicit latest

EVIDENCE / PROVENANCE
Claims · receipts · readbacks · gaps

DATA / CATALOG GOVERNANCE
cross-cutting layer
contracts · metadata · quality

Local execution adapter — historical view
Windows ? WSL2 ? /opt/mm
adapter ? architecture

Bounded scientific modules

MM-ENVIRONMENT
climate · soil
geospatial / GEE

MM-PLANT
DSSAT · forage
plant response

MM-ANIMAL
production · biological
constraints

MM-ECONOMIC
economics · risk
decision outputs

MM-INTEGRATION-CORE
contract-bound
orchestration

MM-DATA-CATALOG
contracts · metadata
provenance

BOUNDARY RULE
module internal state ? inter-module API

Permitted inter-module path

MODULE OUTPUT

VERSIONED DATA CONTRACT

CATALOG RECORD

PROVENANCE / EVIDENCE BINDING

INTEGRATION ADAPTER

CONSUMER MODULE

Evidence-producing development lifecycle · every transition is Gate-governed

1 Scientific need / intended use
Gate
2 Requirements
Gate
3 Architecture & interfaces
Gate
4 Detailed design
Gate
5 Implementation
Gate
6 Verification
Gate
7 Validation
Gate
8 Independent readback / adversarial assurance
Gate
9 Freeze / release candidate

VERIFICATION
Did we build the system correctly?

VALIDATION
Did we build the correct system for intended scientific use?

Independent MM assurance ? formal external IV&V

FROZEN / CERTIFIED ? RUNTIME AUTHORIZED

local verification | primary scientific runtime | oracle runtime | GEE runtime | production runtime